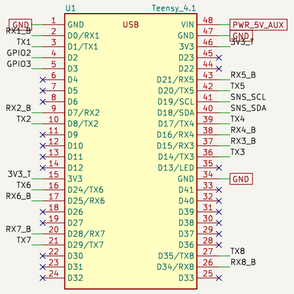
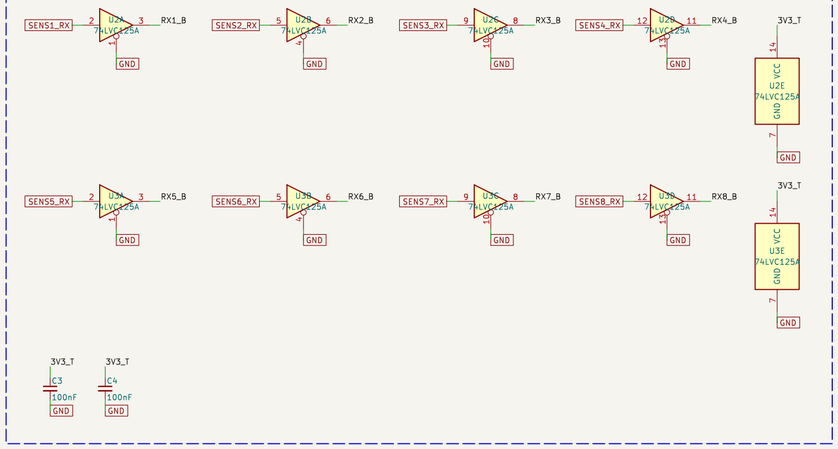
POWER INPUT



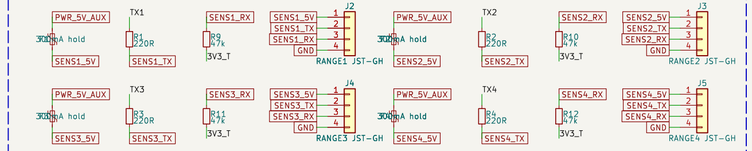
TEENSY 4.1



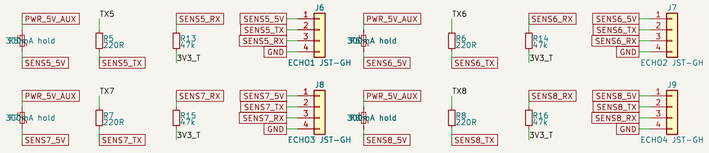
RX LEVEL BUFFERS (5 V-TOLERANT)



SENSOR PORTS 1–4 (RANGEFINDERS)



SENSOR PORTS 5–8 (ECHO SENSORS)



EXPANSION



MECHANICAL



POWER FLAGS (ERC)



NOTES

1. Teensy pins use DIP-48 numbering (1–24 down the left, 25–48 up the right, USB at top). Verify against the PJRC Teensy 4.1 pinout card.
2. Cut the VUSB–VIN trace on the Teensy so the Jetson's USB 5 V and PWR_5V_AUX never connect.
3. U2/U3 use 74LVC125A at 3.3 V: inputs accept 5 V, so each RX line is safe for 3.3 V or 5 V sensors.
4. Trigger/echo sensors use the same ports: TXn becomes TRIG, RXn becomes ECHO.
5. R41/R42 I2C pull-ups on J10 are DNP.

Rev A bootstrap
Bootstrapped by hardware/tools/gen_kicad.py; edit in KiCad from here on
Cut the Teensy VUSB–VIN trace; VIN is fed from J1
Teensy 4.1 carrier: 8 fused UART sensor ports with 5 V–tolerant RX buffers
Jetson drone electronics

Sheet: /
File: sns.kicad_sch

Title: SNS – Sensor Hub

Size: A2	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1